

MEASURING GITHUB COPILOT'S IMPACT ON PRODUCTIVITY

- GitHub Copilot 補全行為 vs 生產力研究
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- 日期：2025/12/09

研究背景與動機

-隨著AI技術的快速發展，像是GitHub Copilot這類AI輔助程式設計工具逐漸成為開發者的常用幫手。

-雖然有人認為這些工具能提升開發效率，但實證研究較少。本報告旨在了解AI工具（特別是Copilot）對開發者 Productivity（生產力）的實際影響。



研究問題

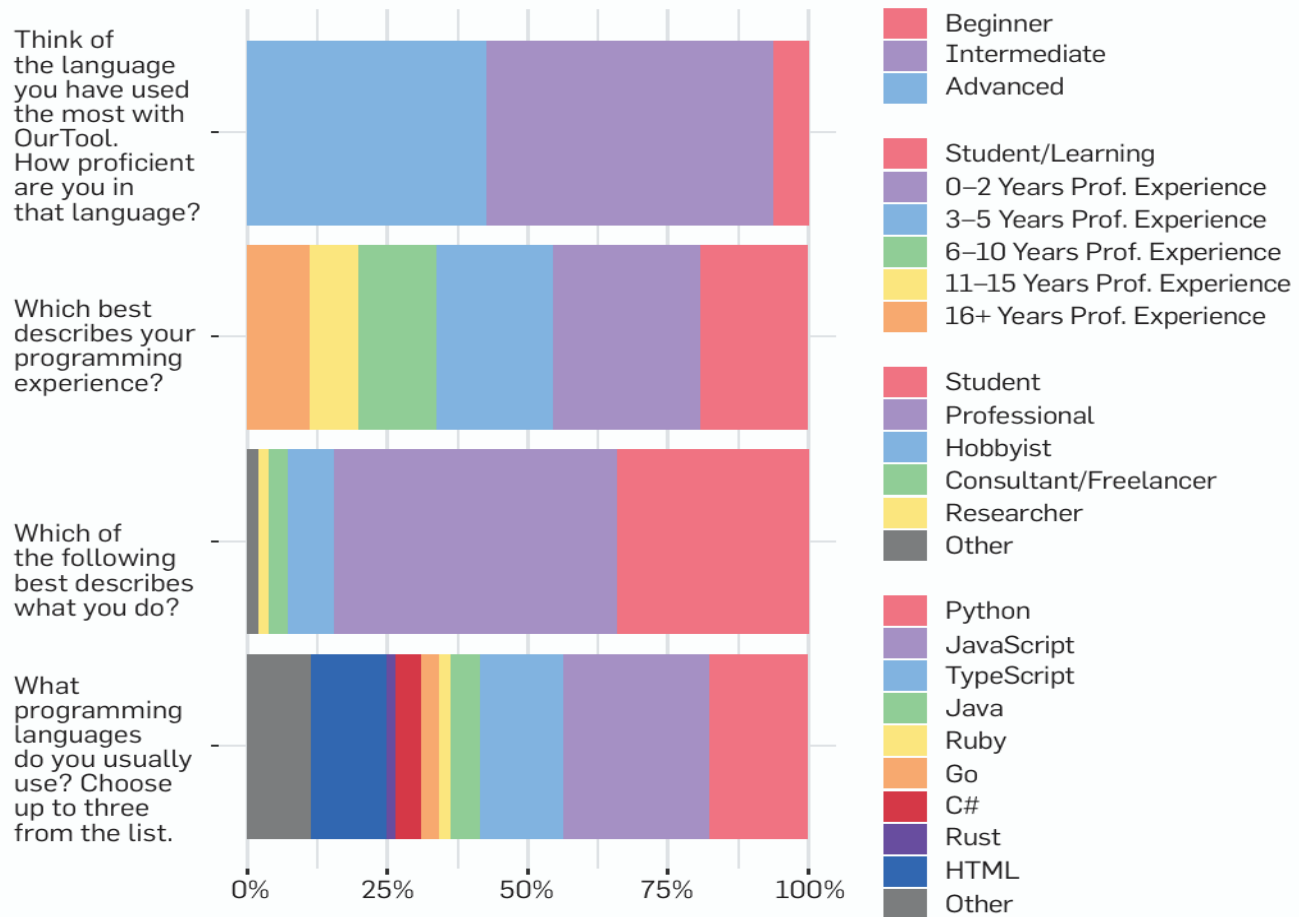
- 1. 哪些補全提升生產力？
- 2. 補全數量是否有用？
- 3. 哪些族群受益？
- 4. 情境是否影響接受程度？

研究方法概述

- - IDE 數據+ 個人體驗問卷
- - 指標：shown/accepted/persistence/contribution
- - 分析：相關/降維/回歸/時間週期

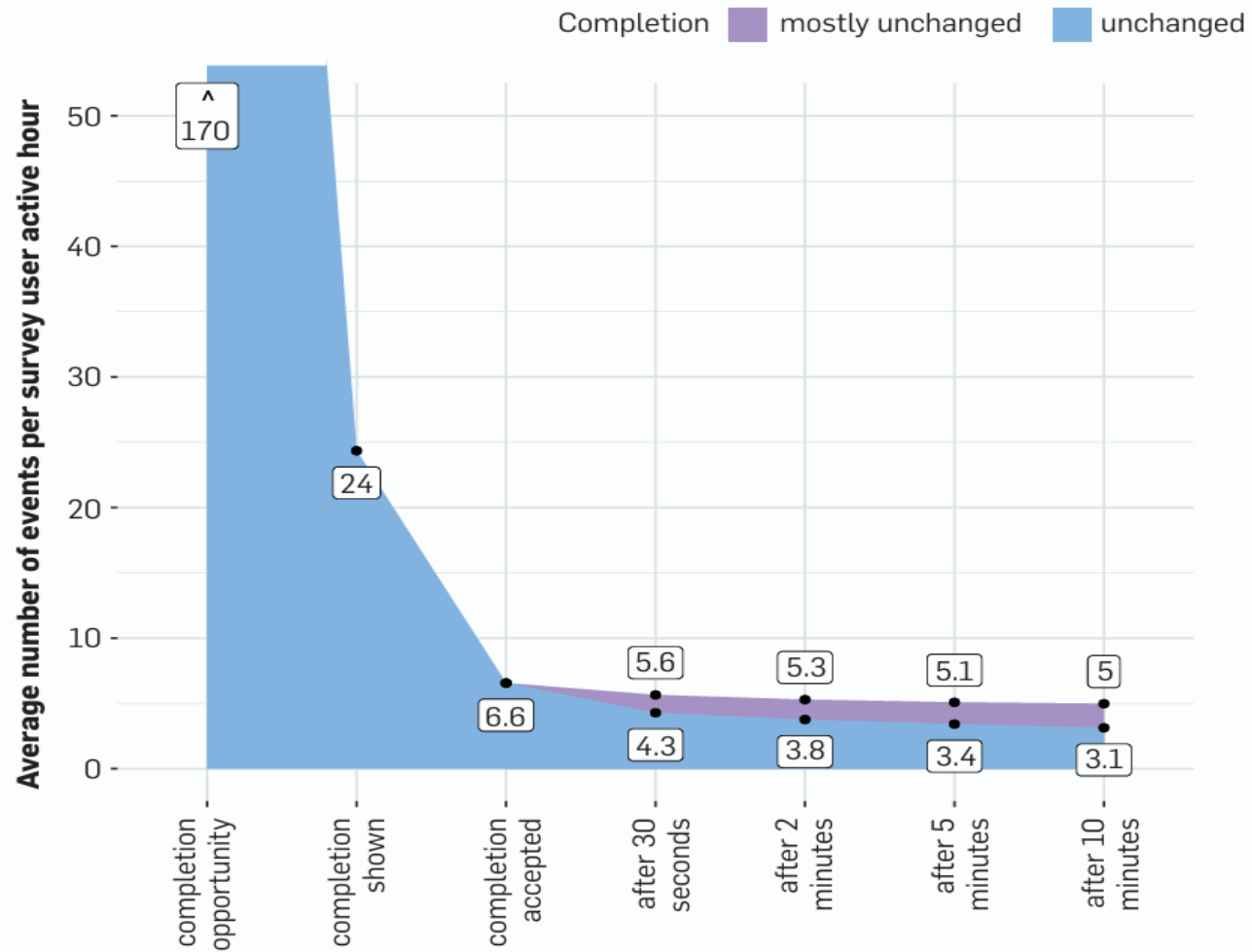
研究對象

Figure 2. Demographic composition of survey respondents.



程式碼補全流程階梯

Figure 1. GitHub Copilot's code completion funnel.



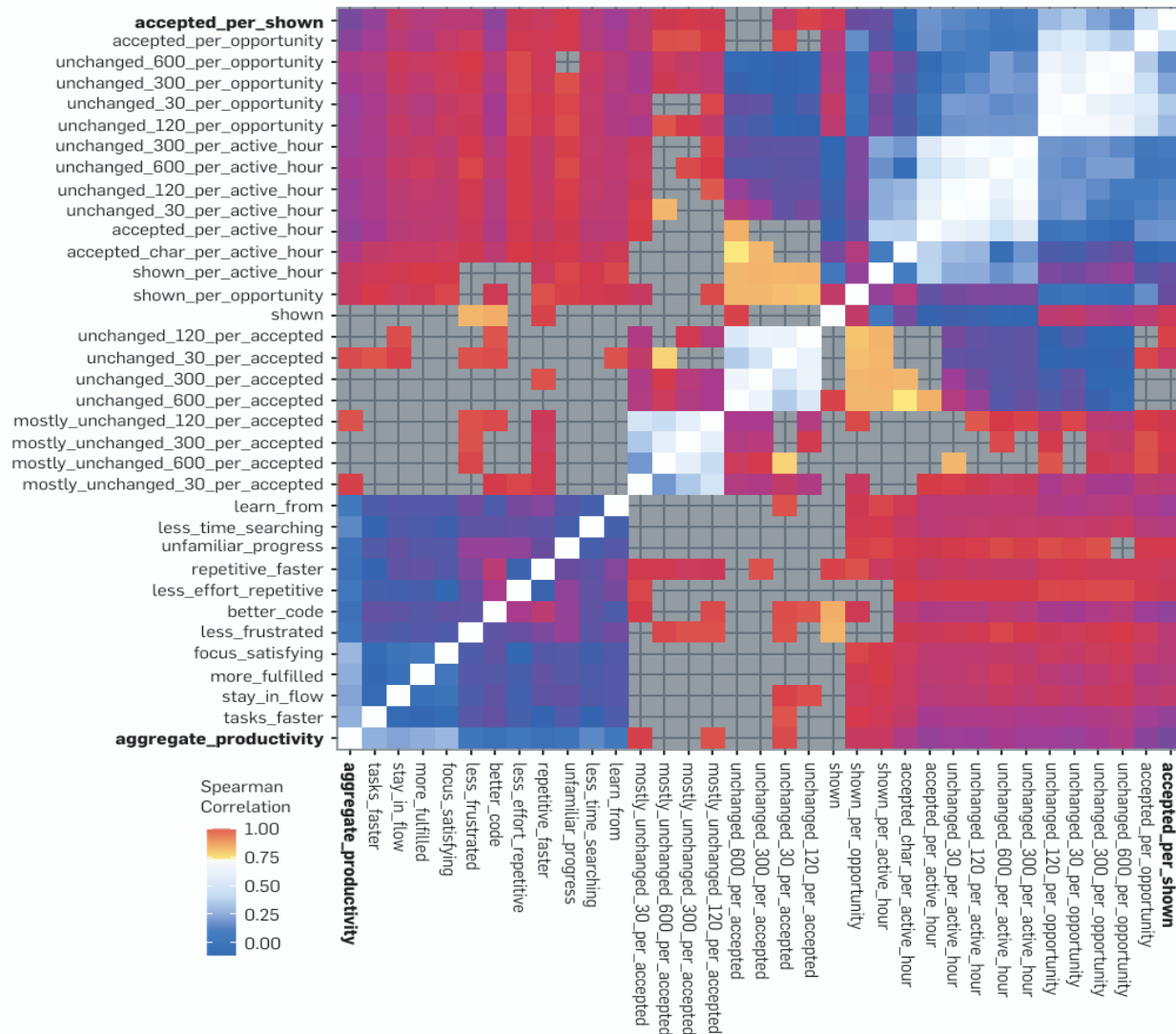
行為指標

Table 2. The core set of measurements considered in this article.

Natural name	Explanation
<u>Shown rate</u>	Ratio of completion opportunities that resulted in a completion being shown to the user
<u>Acceptance rate</u>	Ratio of shown completions accepted by the user
Persistence rate	Ratio of accepted completions unchanged after 30, 120, 300, and 600 seconds
Fuzzy persistence rate	Ratio of accepted completions mostly unchanged after 30, 120, 300, and 600 seconds
Efficiency	Ratio of completion opportunities that resulted in a completion accepted and unchanged after 30, 120, 300, and 600 seconds
<u>Contribution speed</u>	Number of characters in accepted completions per distinct, active hour
<u>Acceptance frequency</u>	Number of accepted completions per distinct, active hour
Persistence frequency	Number of unchanged completions per distinct, active hour
Total volume	Total number of completions shown to the user
Loquaciousness	Number of shown completions per distinct, active hour
Eagerness	Number of shown completions per opportunity

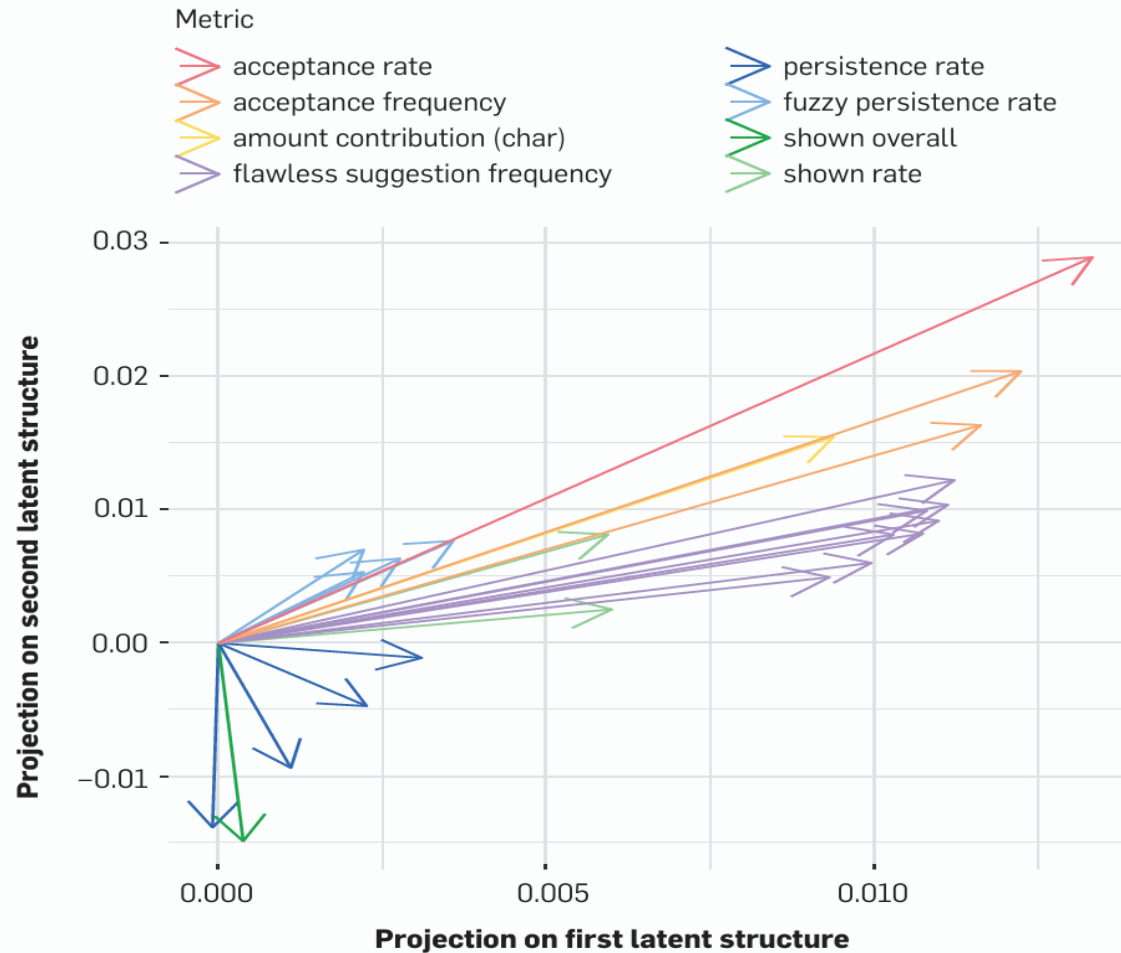
相關矩陣

Figure 3. Correlation between metrics. Metrics are ordered by similarity based on distance in the correlation matrix, except for manually fixing the aggregate productivity and acceptance rate at the end for visibility.



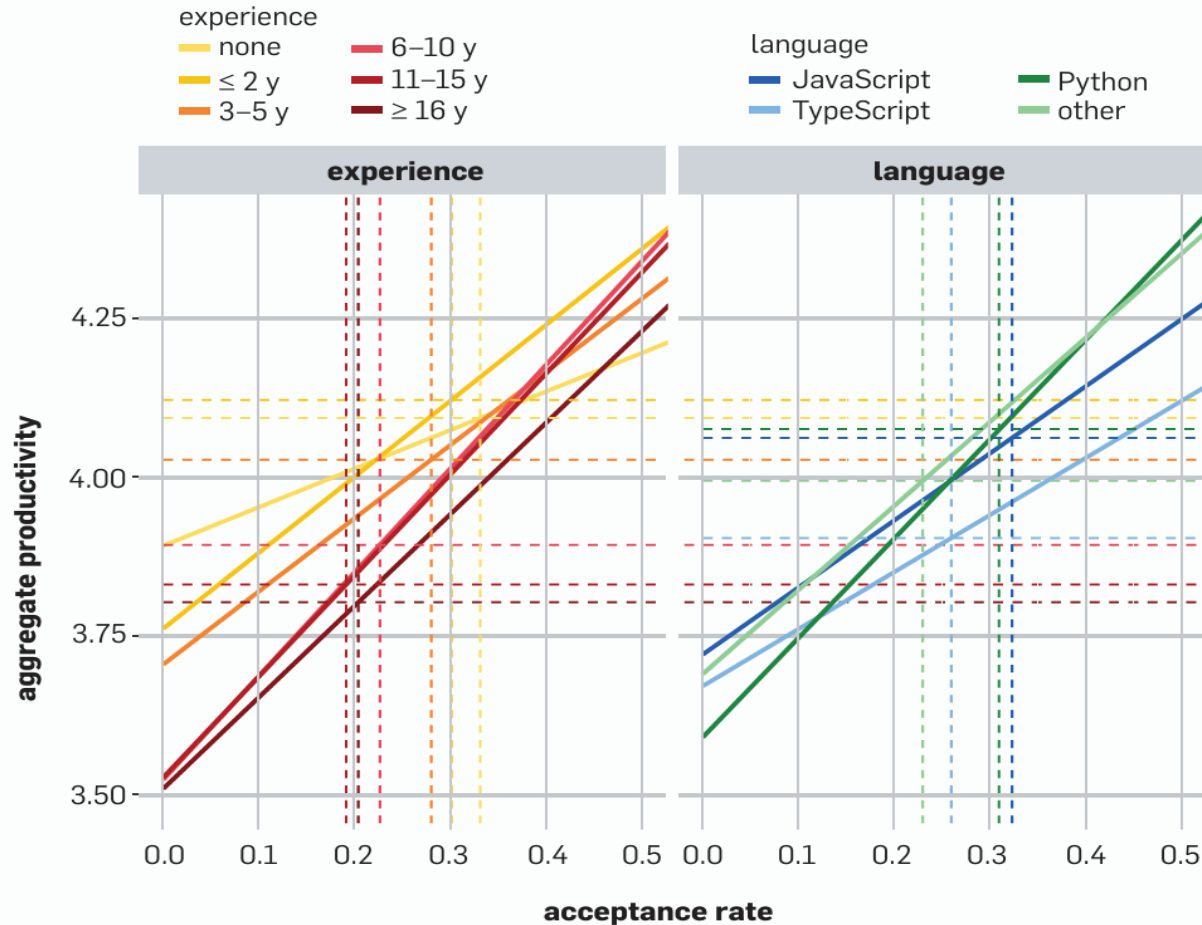
指標降維結果

Figure 4. Different metrics clustering in latent structures predicting perceived productivity. We color the following groups: flawless suggestions (counting the number of unchanged suggestions), persistence rate (ratio of accepted suggestions that are unchanged), and fuzzy persistence rate (accepted suggestions that are mostly unchanged).



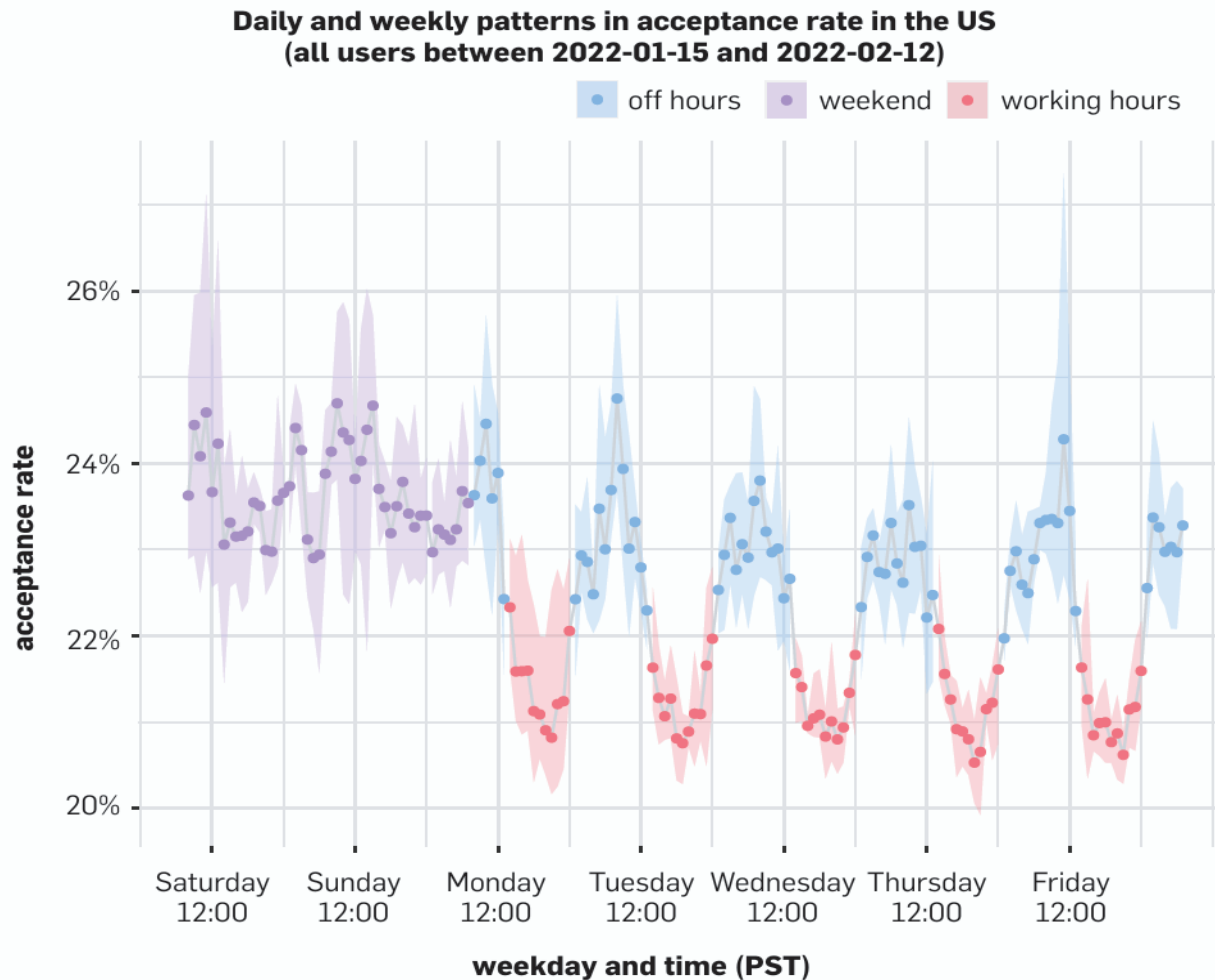
族群差異

Figure 5. Linear regressions between acceptance rate and aggregate productivity by subgroup defined through years of professional experience or programming language use. Dashed lines denote averages. The x-axis is clipped at (0, 0.5), and 95% of respondents fall into that range.



時間週期

Figure 6. Average acceptance rate during the week. Each point represents the average for a one-hour period, whereas the shaded ribbon shows the min-max variation during the observed four-week period.



討論

- 1. 品質比數量重要
- 2. 效果因人而異
- 3. 情境影響接受度

研究貢獻

- 首次明確量化「使用行為」與「生產力感受」之間的連結。
→ Acceptance rate 能預測 perceived productivity。
- 證明補全的「品質」比「數量」更重要。
→ Persistence 不如 acceptance rate 有預測力。
- 揭露不同族群、不同時間的使用差異。
→ 新手效益大、JS/TS 使用者較高、晚上與週末 acceptance rate 上升。

結論

- AI 補全提升屬小而穩定
- 被接受並保留的補全最有價值
- 中階工程師與 JS 使用者最受益